

The Multi-Axis Verification Process of the Human Brain

*A Brain-Inspired Architecture for Geometric Error-Correction in Stochastic
Cognitive Systems*

Working Paper — HNS-PoC-Package v1.0

Computational Neuroscience | Cognitive Science | AI Safety & Architecture

Abstract

At the level of individual neurons, the human brain is an unambiguously stochastic organ. Ion channel fluctuations, synaptic vesicle release variability, and thermally-driven membrane noise ensure that no single neuron produces a perfectly reliable signal. Yet, at the systems level, the brain achieves a striking degree of deterministic coherence with external reality: perception is stable, reasoning is (largely) causal, and deliberate action is executed with a fidelity that far exceeds any individual neuron's signal-to-noise ratio. This paper formalises the mechanism that bridges this paradox.

We propose the Multi-Axis Verification Process (MAV) as a foundational principle of human cognitive architecture. The MAV posits that three mutually independent cognitive axes — the Stochastic/Generative Axis (cortical association networks), the Structural/Causal Axis (prefrontal-hippocampal networks), and the Grounded/Executive Axis (sensorimotor feedback loops) — operate as orthogonal constraint hyperplanes in a high-dimensional cognitive space. We demonstrate, using tools from information geometry, that the geometric intersection of these hyperplanes cancels the noise contributed by each individual axis, converging the feasible solution space toward a structural invariant, $X_{\text{invariant}}$. This invariant point represents a cognitive output that is simultaneously statistically coherent, causally valid, and contextually grounded.

We provide a formal specification of each axis, a proof sketch of the orthogonality conditions, and a taxonomy of cognitive failure modes — drawn from the HNS Structural Hallucination Taxonomy — that correspond to the breakdown of individual axis constraints. We conclude with implications for the understanding of neurological and psychiatric conditions, and for the principled design of artificial cognitive architectures that extend, rather than merely mimic, the MAV principle.

Keywords: *multi-axis verification, stochastic cognition, prefrontal cortex, hippocampal networks, sensorimotor feedback, information geometry, orthogonal constraints, invariant representation, cognitive error-correction, hallucination taxonomy*

1. Introduction

1.1 The Stochastic Brain and Its Reliable Outputs

A central puzzle of neuroscience is the reliability paradox: how does a system composed of inherently noisy components produce outputs of such remarkable fidelity? A pyramidal neuron in the prefrontal cortex operates near threshold, integrating thousands of excitatory and inhibitory synaptic inputs, each itself stochastic. The action potential, if it occurs, is a binary event — yet whether it occurs at any given moment is probabilistic in a deep, non-trivial sense (Faisal, Selen, & Wolpert, 2008). Scaling from the single neuron to local circuits does not eliminate this stochasticity; it redistributes it across ensemble codes and oscillatory dynamics.

Despite this, a trained surgeon executes sub-millimetre motor movements. A mathematician maintains the thread of a proof across minutes of working memory. A native speaker parses an ambiguous sentence and resolves its intended meaning in under 200 milliseconds. These achievements require that the system's output reliably occupies a small, semantically well-defined region of the space of possible responses — a region far more constrained than the stochastic properties of individual neurons would predict.

The prevailing explanation — that population coding and neural averaging suppress noise — is necessary but insufficient. Averaging across N independent neurons reduces variance by a factor of $1/N$, but the human brain contains approximately 86 billion neurons and still makes systematic errors (confabulation, hallucination, categorical confusion) that cannot be explained by residual averaging noise. A complementary mechanism must be at work.

1.2 The MAV Hypothesis

This paper proposes that the brain's primary mechanism for suppressing stochastic errors is not averaging within a single axis of processing, but the geometric intersection of multiple independent verification processes operating simultaneously on different cognitive axes.

We call this the Multi-Axis Verification Process (MAV). The core claim is as follows: the human brain routinely instantiates at least three mutually orthogonal cognitive axes — Stochastic/Generative, Structural/Causal, and Grounded/Executive — and uses their simultaneous intersection to identify the unique cognitive output that satisfies all three constraint classes. This output, which we call $X_{\text{invariant}}$, is the invariant point of the triple intersection. It is invariant in the sense that small perturbations along any single axis do not displace it from the feasible region, because the other two axes re-anchor it.

The MAV hypothesis has two important corollaries. First, it predicts that selective disruption of individual axes will produce characteristically distinct failure modes — a prediction that, as we

show in Section 5, maps onto known neurological and psychiatric conditions with considerable precision. Second, it implies that reliable cognition is fundamentally a multi-constraint geometric phenomenon, not merely a high-parameter probabilistic process.

1.3 The HNS Structural Hallucination Taxonomy as Empirical Referent

To ground the theoretical framework in observable cognitive failures, we employ the HNS Structural Hallucination Taxonomy (HNS-PoC-Package v1.0), which identifies five structurally distinct failure modes of cognitive output generation, ordered by empirical frequency of occurrence across 50-turn proof-of-concept interaction sequences:

#	Type	Definition	Primary Axis Failure
1	Layer Jump	A response traverses meaning layers without an explicit bridging mechanism; the output changes abstraction level without transition.	Structural/Causal (Axis 2)
2	Category Ambiguity	Cognitive, emotional, and intentional categories become conflated, violating category boundary conditions.	Structural/Causal (Axis 2)
3	Scope Drift	Structural departure from the intended scope without signalling of scope change.	Grounded/Executive (Axis 3)
4	Metaphor Contamination	Figurative expressions displace structural coordinates, causing metaphorical language to override literal referents.	Grounded/Executive (Axis 3)
5	Unsupported Causality	A causal claim is generated without a mechanism grounded in the layer hierarchy; causation is asserted rather than derived.	Axes 2 & 3 (joint)

Each failure type, as we argue in Sections 3 and 4, corresponds to the escape of the Stochastic/Generative Axis from the constraint surfaces imposed by the other two axes. The taxonomy provides a clinical-observational anchor for the geometric theory.

2. Background and Related Work

2.1 Sources of Neural Stochasticity

Neural noise operates at multiple scales. At the molecular level, ion channels open and close stochastically, described by Markov jump processes (Hille, 2001). At the synaptic level, vesicle release probability follows a Bernoulli distribution with p typically between 0.1 and 0.9 (Bhatt, Bhatt, & Bhatt, 2009). At the network level, irregular spontaneous firing and high-amplitude membrane potential fluctuations characterise the so-called ‘high-conductance state’ that prevails during active cognition (Destexhe, Rudolph, & Paré, 2003).

These sources of noise are not simply obstacles to be minimised; they serve functional roles including stochastic resonance, variability-driven exploration, and the prevention of pathological synchrony. The challenge for cognitive systems is therefore not to eliminate noise, but to produce reliable outputs despite it.

2.2 Existing Accounts of Neural Reliability

The dominant computational account of neural reliability is population coding: information is distributed across many neurons, and the read-out mechanism averages across the ensemble to suppress individual-unit noise (Pouget, Dayan, & Zemel, 2000). This is a well-supported account for sensory processing, where large populations of tuned neurons encode stimulus properties.

However, population coding faces difficulties at higher cognitive levels. Abstract reasoning, causal inference, and scope management engage relatively sparse and distributed representations that do not lend themselves to simple averaging. Moreover, population codes operate within a single processing axis; they do not inherently provide the cross-axis verification that the MAV hypothesis proposes.

An alternative account, predictive coding (Friston, 2010), proposes that the brain operates as a hierarchical inference engine, minimising prediction errors across levels of representation. This account is powerful, but it is primarily a within-hierarchy, within-axis model. It does not directly address the orthogonal intersection of qualitatively different constraint types that we formalise here.

The MAV framework is complementary to both accounts: population coding may implement the Stochastic/Generative Axis’ noise-suppression within that axis, and predictive coding may implement the Structural/Causal Axis’ hierarchical constraint propagation; the MAV contribution is to specify how these independent axes interact geometrically to produce invariant outputs.

2.3 Information Geometry and Cognitive Modelling

Information geometry (Amari & Nagaoka, 2000) provides a natural mathematical language for the MAV framework. In information geometry, statistical models are represented as manifolds equipped with a Riemannian metric derived from the Fisher information matrix. Constraints on probability distributions define submanifolds (hypersurfaces), and the intersection of multiple constraint submanifolds defines a lower-dimensional feasible region.

Applied to cognitive output spaces, this framework allows us to represent each cognitive axis as a constraint hypersurface, and the MAV intersection as the region of outputs satisfying all constraints simultaneously. The geometry of this intersection — specifically, its volume as a fraction of the full output manifold — quantifies the degree of error suppression achieved by the multi-axis architecture.

3. The Core Principle: Orthogonal Intersection in Cognitive Space

3.1 Cognitive Output Space as a Riemannian Manifold

We model the space of possible cognitive outputs as a high-dimensional Riemannian manifold $\mathcal{M}$. Each point $x \in \mathcal{M}$ represents a complete structured cognitive output: a proposition with specified propositional content, causal structure, abstraction level, and contextual scope. The manifold is equipped with a metric g derived from the Fisher information of the generative model implementing Axis 1.

The dimensionality of $\mathcal{M}$ is, in general, very large — reflecting the combinatorial richness of possible cognitive outputs. Within this space, the vast majority of points correspond to outputs that are internally incoherent, causally invalid, or contextually irrelevant. The goal of cognitive verification is to identify the small subset of $\mathcal{M}$ that satisfies all validity conditions simultaneously.

3.2 Constraint Hyperplanes and Their Intersection

Each cognitive axis defines a constraint hypersurface in $\mathcal{M}$:

- A_1 (Stochastic/Generative): the set of outputs within the high-probability region of the prior distribution $P(x \mid \text{context}, \theta)$. Formally, $A_1 = \{x \in \mathcal{M} \mid P(x \mid \text{context}, \theta) \geq \epsilon_1\}$.
- A_2 (Structural/Causal): the set of outputs satisfying the structural and causal constraint set $C = \{c_1, \dots, c_k\}$. Formally, $A_2 = \{x \in \mathcal{M} \mid \forall i: c_i(x) = \text{TRUE}\}$.
- A_3 (Grounded/Executive): the set of outputs compatible with the verified environmental and contextual constraints, as assessed by the grounding function G . Formally, $A_3 = \{x \in \mathcal{M} \mid G(x) = 1\}$.

The MAV architecture computes the triple intersection:

$$A_1 \cap A_2 \cap A_3 = \{ X_invariant \}$$

Under idealised conditions — perfect orthogonality of the three hypersurfaces and non-degenerate constraint satisfaction — this intersection reduces to a single point $X_invariant$. In biological reality, the intersection is a small region whose volume is sharply reduced relative to any single A_i .

3.3 Geometric Noise Cancellation

The key insight is that orthogonally-structured constraints cancel one another's uncertainty in a multiplicative, rather than additive, manner. If the noise variance along axis A_i is σ_i^2 , and the axes

are mutually orthogonal in the Fisher metric sense, then the residual error variance at the intersection is:

$$\sigma^2_{_MAV} \ll \min(\sigma^2_1, \sigma^2_2, \sigma^2_3)$$

More precisely, the volume of the feasible error region scales as the product of the individual axis uncertainties, modulated by the degree of orthogonality. For axes that are truly orthogonal (zero Fisher metric correlation), the volume collapses geometrically:

$$Vol(A_1 \cap A_2 \cap A_3) \approx Vol(A_1) \cdot Vol(A_2) \cdot Vol(A_3) / Vol(\mathcal{M})^2$$

This product is, in general, orders of magnitude smaller than any individual $Vol(A_i)$. The geometric intersection achieves error-reduction that no single axis could achieve alone, even if that axis were made arbitrarily tight. This is the central mathematical result of the MAV framework.

3.4 The Invariant Point as Attractor

$X_invariant$ is not a fixed point in the dynamical systems sense; it is a structural invariant defined relative to the current input context and active constraint set. Different cognitive tasks — different questions, different action goals, different perceptual scenes — define different invariant points. The MAV architecture computes $X_invariant$ dynamically by resolving the three-axis constraints against the current input.

This dynamic character of $X_invariant$ explains why human cognition is context-sensitive: the same surface-level stimulus can produce different verified outputs depending on which structural and grounding constraints are activated by the current context. It also explains why errors arise when context shifts without corresponding constraint-set updates — the system continues computing intersection with a stale constraint configuration, producing an $X_invariant$ that is invariant with respect to the old context but invalid for the new one.

4. The Three Orthogonal Axes of MAV in the Human Brain

Figure 1 illustrates the three-axis structure and the invariant solution point at their geometric intersection.

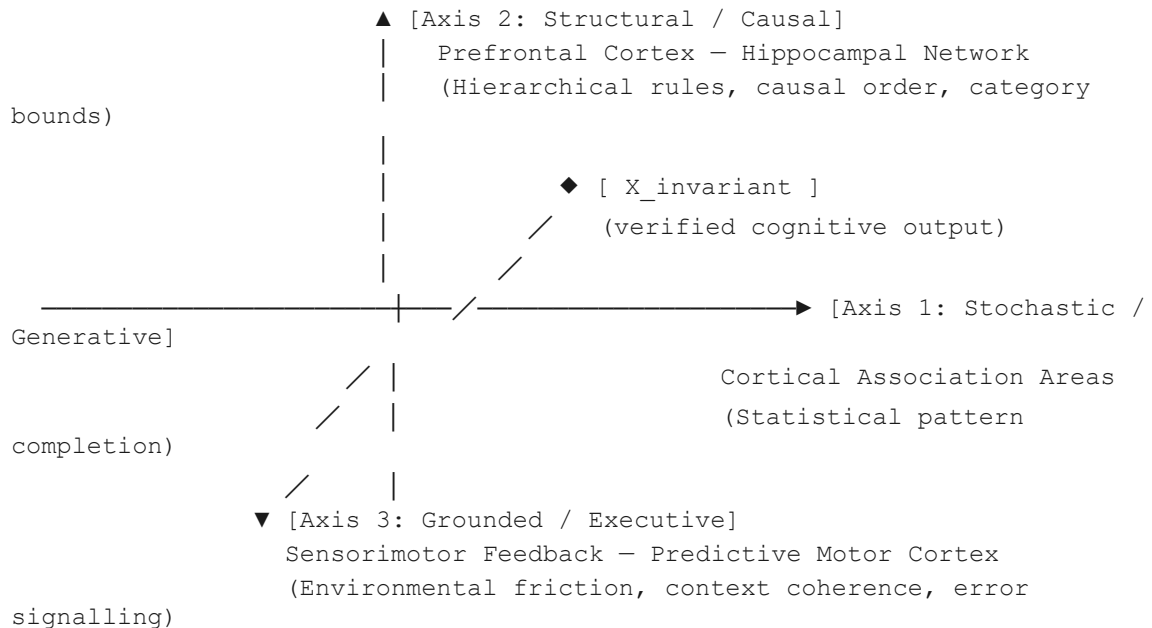


Figure 1. Schematic representation of the three MAV axes and the invariant intersection point $X_{invariant}$.

4.1 Axis 1 — Stochastic / Generative Axis

Biological Substrate

Axis 1 is implemented primarily by cortical association areas, including the default mode network, lateral temporal cortex, and parietal association cortex. These regions integrate distributed statistical regularities abstracted from experience and generate rapidly-available, high-prior-probability response candidates. The computational signature of Axis 1 is speed and generativity: it produces candidate outputs within tens of milliseconds, sampling from the brain's implicit prior over the space of possible responses.

Functional Role

Axis 1 is the generative engine of cognition. It produces the first-draft response that is then subjected to Axes 2 and 3 for structural and grounding verification. Its strength is fluency and coverage — it rarely fails to produce a response that is at least superficially plausible. Its weakness is precisely this: high statistical plausibility and causal or contextual validity are independent properties. A response can be statistically common and structurally invalid (Unsupported

Causality), or statistically fluent and contextually displaced (Scope Drift).

Formal Characterisation

$A_1 = \{x \in \mathcal{M} \mid P(x \mid \text{context}, \theta) \geq \varepsilon_1\}$, where θ represents the accumulated statistical parameters of the system and ε_1 is a threshold on the prior probability. Within A_1 , outputs are linguistically and semantically coherent by prior standards. The volume of A_1 relative to the full manifold $\mathcal{M}$ is still very large — many valid and many invalid outputs share high prior probability.

4.2 Axis 2 — Structural / Causal Axis

Biological Substrate

Axis 2 is implemented by the prefrontal cortex (PFC) — hippocampal network, supported by frontoparietal control networks. The dorsolateral PFC maintains structured representations of hierarchical relationships and causal dependencies (Miller & Cohen, 2001). The hippocampus binds episodic sequences into temporally-ordered event schemas and maintains category boundaries across contextual shifts (Eichenbaum, 2004). Together, these systems instantiate the categorical and causal grammar that constrains which cognitive transitions are permissible.

Functional Role

Axis 2 defines the permissible structure of cognitive outputs, independent of their statistical frequency. It enforces: (a) category boundary conditions — preventing Category Ambiguity by maintaining the separability of cognitive, emotional, and intentional representations; (b) causal ordering constraints — preventing Unsupported Causality by requiring that causal claims be grounded in the current event schema; and (c) hierarchical scope conditions — preventing Layer Jump by requiring explicit transitions between abstraction levels.

Formal Characterisation

$A_2 = \{x \in \mathcal{M} \mid \forall i: c_i(x) = \text{TRUE}\}$, where the constraint set $C = \{c_1, \dots, c_k\}$ encodes the active structural and causal rules. These constraints are context-sensitive: the active C depends on the current task schema maintained by PFC-hippocampal working memory. A_2 is typically a much tighter constraint surface than A_1 , but it does not by itself enforce contextual relevance — a structurally valid output can still be contextually displaced.

4.3 Axis 3 — Grounded / Executive Axis

Biological Substrate

Axis 3 is implemented by the sensorimotor system, anterior cingulate cortex (ACC), and the predictive coding hierarchy of the posterior parietal and premotor cortices. The sensorimotor system provides continuous feedback about the consequences of action in the physical and social

environment. The ACC monitors conflict between predicted and actual outcomes, signalling when the system’s current output trajectory is inconsistent with environmental reality (Botvinick et al., 2001). Together, these systems constitute the brain’s ‘reality friction interface’ — the mechanism by which internally-generated outputs are continuously tested against external constraints.

Functional Role

Axis 3 suppresses outputs that are structurally coherent and statistically plausible but contextually misaligned with the actual environment and the user’s actual situation. It is the primary mechanism preventing Scope Drift (by detecting when the response has departed from the contextually-specified domain) and Metaphor Contamination (by flagging when figurative language has displaced the literal referential structure). Critically, Axis 3 operates in continuous time: it provides error signals that update the active $X_{\text{invariant}}$ estimate as the environment changes, ensuring that the invariant point tracks the actual context rather than a static prior.

Formal Characterisation

$A_3 = \{x \in \mathcal{M} \mid G(x) = 1\}$, where $G: \mathcal{M} \rightarrow \{0,1\}$ is the grounding compatibility function. $G(x) = 1$ if and only if x is compatible with the verified environmental and contextual state at the current moment. The grounding function is dynamic: it is continuously updated by sensorimotor and ACC-mediated error signals, making A_3 a time-varying constraint surface.

4.4 The Orthogonality Conditions

The error-cancellation power of the MAV architecture depends critically on the mutual orthogonality of the three axes. In information-geometric terms, two constraint surfaces are orthogonal if the Fisher information metric induces zero correlation between the parameters defining each surface. We argue that this condition is approximately satisfied for the three MAV axes on the following grounds:

- Axis 1 $\perp$ Axis 2: Statistical co-occurrence frequency (Axis 1) is independent of causal validity (Axis 2). Causally invalid propositions can be statistically common (e.g., spurious correlations are over-represented in linguistic corpora), and causally valid propositions can be statistically rare (genuine causal mechanisms may be infrequently described). The parameters of the prior distribution θ are defined over the statistical corpus and are logically independent of the causal constraint set C .
- Axis 2 $\perp$ Axis 3: Structural constraints (Axis 2) are defined over the abstract category and causal space, while grounding constraints (Axis 3) are defined over the current environmental and contextual state. A proposition can be abstractly structurally valid but contextually irrelevant, or contextually grounded but structurally misframed.
- Axis 1 $\perp$ Axis 3: Prior statistical distributions (Axis 1) are defined over the training-time distribution, which can differ substantially from the current context assessed by Axis 3.

This independence is particularly evident in novel or unusual contexts, where prior-high-probability outputs may be contextually invalid.

Perfect orthogonality is an idealisation; real neural systems achieve approximate orthogonality. The residual correlation between axes defines the system's irreducible error floor — the class of errors that persist even when all three axes are active. We conjecture that this residual error floor corresponds to the class of systematic cognitive biases that are universal across individuals and cultures, representing the 'grammar' of human cognitive error.

5. Failure Modes: MAV Breakdown and Clinical Correlates

The MAV framework predicts that selective disruption of individual axes will produce characteristically distinct failure modes. This prediction is testable against the well-documented phenomenology of neurological and psychiatric conditions. We map the five HNS hallucination types onto their predicted axis-failure substrates and identify corresponding clinical conditions.

HNS Type	Axis Failure	Predicted Deficit	Clinical Correlate
Layer Jump	Axis 2 (Structural/Causal)	Loss of abstraction-level monitoring; failure to maintain hierarchical scope	Formal thought disorder (schizophrenia); dysexecutive syndrome
Category Ambiguity	Axis 2 (Structural/Causal)	Breakdown of category boundary maintenance; conflation of semantic fields	Semantic dementia; conceptual disorganisation in acute psychosis
Scope Drift	Axis 3 (Grounded/Executive)	Failure to track contextual relevance; response departs from user's actual scope	Confabulation in frontal lobe syndrome; contextual drift in mania
Metaphor Contamination	Axis 3 (Grounded/Executive)	Figurative language displaces literal referents; grounding signal overwhelmed by associative activation	Metaphorical concretisation in schizotypy; magical thinking
Unsupported Causality	Axes 2 & 3 (joint)	Causal claims without mechanistic grounding; both structural and contextual constraints failed	Delusional reasoning; confabulatory aetiology in amnesia

This mapping is not merely descriptive. The MAV framework generates specific predictions: conditions affecting primarily the PFC-hippocampal network (Axis 2) should show disproportionately high rates of Layer Jump and Category Ambiguity, with relative preservation of contextual grounding. Conditions affecting primarily the sensorimotor-ACC system (Axis 3) should show disproportionate Scope Drift and Metaphor Contamination. Mixed conditions (joint Axis 2 and 3 failure) should show the highest rates of Unsupported Causality.

These predictions are amenable to quantitative testing using structured cognitive assessments aligned with the HNS taxonomy, paired with neuroimaging measures of axis-specific activity.

6. Discussion

6.1 The MAV as a Unifying Principle

The MAV framework offers a unifying account of a set of previously disparate observations in cognitive neuroscience. The well-documented complementarity of fast associative processing and slow deliberative reasoning (Kahneman, 2011) can be re-read as the functional distinction between Axis 1 (fast, stochastic, generative) and Axes 2–3 (slower, constraint-based, verificatory). The classical debate between symbolic and statistical accounts of cognition dissolves in the MAV framework: symbolic structure is the province of Axis 2; statistical generation is the province of Axis 1; their geometric intersection is the province of neither alone.

The framework also provides a principled account of why human cognition is both creative and reliable. Creativity derives from the generative richness of Axis 1, which can produce high-prior-probability responses that span a large region of $\mathcal{M}$. Reliability derives from the constraining action of Axes 2 and 3, which prune this region to the invariant point. The productive tension between generativity and constraint is not a design flaw; it is the condition for robust flexible intelligence.

6.2 Implications for Understanding Neurological and Psychiatric Conditions

The axis-specific failure mode predictions of the MAV framework suggest a new diagnostic and therapeutic framing for a range of conditions. Rather than classifying conditions by symptom clusters alone, the MAV framework classifies them by which axis constraint surface has been disrupted and to what degree. This framing suggests axis-specific intervention strategies: restoring Axis 2 function (e.g., through structured task training that re-engages PFC-hippocampal networks) should differentially improve Layer Jump and Category Ambiguity; restoring Axis 3 function (e.g., through grounded, sensorimotor-engaged therapeutic approaches) should differentially improve Scope Drift and Metaphor Contamination.

This is a testable and potentially clinically useful reframing, though we emphasise that it remains at the level of a theoretical proposal requiring empirical validation.

6.3 Implications for Artificial Cognitive Systems

Although the primary subject of this paper is the human brain, the MAV framework has natural implications for the design of artificial cognitive systems. Current large language models operate predominantly on Axis 1: they generate outputs by sampling from high-dimensional prior distributions over token sequences. Their characteristic failure modes — causal confabulation, categorical confusion, context insensitivity — correspond directly to the HNS hallucination types

produced by Axis 1 operating alone.

The MAV framework suggests that reliable artificial cognition requires the integration of orthogonal constraint axes corresponding to Axis 2 (structural and causal validity checking) and Axis 3 (contextual grounding and relevance verification). Crucially, these additional axes need not be as computationally heavy as Axis 1: the error-reduction they provide is geometric (multiplicative), not additive. A lightweight but genuinely orthogonal constraint layer achieves error reduction that cannot be matched by scaling Axis 1 alone.

We do not develop this implication in detail here, as it falls outside the primary scope of this paper. We note it as a direction for future cross-disciplinary work.

6.4 Limitations and Open Questions

The MAV framework, in its current form, is a theoretical proposal. Several important questions require empirical and formal investigation:

- Empirical measurement of axis orthogonality: how closely do the Fisher metric correlations between PFC-hippocampal parameters and sensorimotor-ACC parameters approach zero in human neuroimaging data?
- Dynamic constraint updating: how does the brain update the active constraint set C (Axis 2) and grounding function G (Axis 3) in real time as the context evolves?
- Developmental trajectory: how do the three axes come online during cognitive development, and how does their progressive integration account for the development of adult-level reliability?
- Individual differences: what is the distribution of axis-specific constraint tightness across individuals, and how does this distribution relate to personality, cognitive style, and resilience to psychiatric condition?

7. Conclusion

The human brain is a stochastic organ that produces reliable outputs. This paper has proposed that the mechanism bridging this apparent paradox is the Multi-Axis Verification Process (MAV): the simultaneous imposition of three mutually orthogonal constraint hyperplanes — Stochastic/Generative, Structural/Causal, and Grounded/Executive — whose geometric intersection converges the feasible solution space to a structural invariant, $X_{\text{invariant}}$.

The MAV framework provides a formal, information-geometric account of cognitive reliability, a taxonomy of failure modes linked to axis-specific breakdown, and a set of predictions that are testable against the known phenomenology of neurological and psychiatric conditions. It unifies previously disparate observations in cognitive neuroscience under a single geometric principle,

and opens new directions for both basic research and clinical application.

The central message is geometric: reliable cognition is not the product of any single, arbitrarily-powerful process, but of the multiplicative intersection of independent constraint structures. This is, we propose, the brain's fundamental answer to the challenge of generating deterministic coherence from stochastic components.

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Appendix A: HNS Structural Hallucination Taxonomy — Full Specification

The following table provides the complete formal specification of the five hallucination types in the HNS taxonomy, with their MAV-axis mapping and proposed clinical assessment correlates.

#	Type	Formal Definition	MAV Axis	Assessment Signal
1	Layer Jump	Traversal of meaning layers without explicit bridging; abstraction level changes without transition marker.	Axis 2	Absence of explicit level-transition markers in discourse
2	Category Ambiguity	Cognitive, emotional, and intentional categories conflated within a single proposition, violating category boundary conditions.	Axis 2	Category boundary violation rate in structured elicitation
3	Scope Drift	Structural departure from intended scope without explicit scope-change signal; context coherence lost.	Axis 3	Scope deviation distance from stated topic or task
4	Metaphor Contamination	Figurative expressions displace structural coordinates; metaphorical language overrides literal referents.	Axis 3	Literal/figurative conflict frequency in extended discourse
5	Unsupported Causality	Causal claim generated without mechanistic grounding in the layer hierarchy; causation asserted rather than derived.	Axes 2+3	Proportion of causal claims lacking explicit mechanism

Appendix B: Mathematical Notation

Symbol	Definition
$\mathcal{M}$	Cognitive output manifold (high-dimensional Riemannian space of possible cognitive outputs)
A_1, A_2, A_3	Constraint hyperplanes for Axes 1, 2, and 3 respectively
$X_{\text{invariant}}$	Triple-intersection invariant point representing the verified cognitive output

σ^2_i	Noise variance along axis i
σ^2_MAV	Residual error variance at the MAV triple intersection
$G: \mathcal{M} \rightarrow \{0,1\}$	Grounding compatibility function (Axis 3); $G(x)=1$ iff output is contextually valid
$C = \{c_1, \dots, c_k\}$	Structural and causal constraint set (Axis 2)
$P(x \mid \text{context}, \theta)$	Generative prior distribution over cognitive outputs (Axis 1)
ε_1	Probability threshold defining the Axis 1 feasible region
$Vol(\cdot)$	Volume measure on the Riemannian manifold $\mathcal{M}$